Sovereign English Mitra · The ten-day sprint, 27 Oct → 5 Nov 2026

Mitra's English voice moves from Gemini Live to our own native full-duplex core (PersonaPlex 7B, Moshi family) on the two H200s in the office server, trained by us, with the Gemini key removed from the English path on day 9. Four tracks, three gates, one buffer day. Everything else in Mitra (router, SQL, guard, read-back, pipeline, every suite) stays as it is; Kannada stays on Gemini until its own programme.

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Government of Karnataka
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The sprint in numbers

- **10 days** — **Mon 27 Oct → Wed 5 Nov** · six people on four tracks, plus AI ops and legal
- **2 × H200** — **GPU 1** serves the core all day · **GPU 2** brain by day, trainer and factory by night
- **3 gates** — **day 3** a sovereign turn · **day 5** suites green · **day 8** atlas + panel; each with a written fallback
- **< 0.5 s** — **first word** · barge-in cut < 0.25 s · 0 id or number deviations
- **0** — **Gemini calls** on the English path at the rehearsal; nothing from Gemini is ever stored or trained on
- **2 rounds** — **LoRA training** overnight on day 4 and day 7, each judged the next morning

TEN DAYS, FOUR TRACKS

amber columns are gate days · the bottom rows show which card is busy at night and what "done" means each evening

track ↓ · day →	Day 0 Mon 27 Oct Pull everything, measure Gemini	Day 1 Tue 28 Oct The core speaks on our cards	Day 2 Wed 29 Oct Results spoken in her voice	Day 3 Thu 30 Oct Go / no-go on text forcing GATE A	Day 4 Fri 31 Oct Round one trains overnight	Day 5 Sat 1 Nov Round one judged · wake word GATE B	Day 6 Sun 2 Nov Many callers at once	Day 7 Mon 3 Nov Round two trains · regression	Day 8 Tue 4 Nov Freeze GATE C	Day 9 Wed 5 Nov Rehearsal and sign-off
Core & serving 2 people · PersonaPlex, Moshi, text forcing, LoRA, batching	**Read & decide** PersonaPlex and moshi-finetune code read; text-forcing approach on paper; licence read with legal	**Core up** PersonaPlex on GPU 1 (Rust server); Kyutai STT on the user channel; gpt-oss-120b on vLLM, GPU 2	**Text forcing** Function result words forced into the core's text stream, spoken while listening; voice prompt from the consented clip	**Go / no-go** Forcing made clean, or the same-voice TTS fallback wired; latency traced per stage; echo gate tuned	**LoRA round 1** 20:00 on GPU 2: persona, Sakala wording, ids digit by digit, lead phrase while a function runs	**Round 1 judged** Atlas vs the base; pacing and pause fixes; better weights promoted to GPU 1	**Load test** Batched sessions; 200 virtual callers in steps of 25; sessions per card at a held p99 tick; 2-h soak at 50	**LoRA round 2** 20:00: day-5 findings, interruptions and repairs, noisy numbers, officer and CS registers	**Freeze** Round 2 judged; winning weights frozen and hashed; serving config frozen	**Rehearsal** Demo on the two cards, Gemini key removed from the English path; egress log captured
Relay & client 1 person · /ws/live adapter, dispatch, captions, desks, phone	**Prepare** Adapter contract agreed with CS; Gemini adapter isolated behind a switch	**New upstream** PCM in / PCM out behind `/ws/live`; core text as captions; STT text as the user transcript	**Dispatch** STT text → existing brain → `tools.execute` → prefill; read-back question and confirm bar	**Desks alive** Captions word by word, stage reveal, number checks, notifications in session, three desks + portal	**Phone path** Phone layout, Standby and 20 s idle, mute = mic only, Stop = the cut, document reply by the core	**"Hey Mitra"** openWakeWord model trained from Piper clips; no Gemini transcription of room talk	**Scheduler lite** Admission control on the two cards; per-card session count; drain before a weight swap	**Full regression** settle 12 · mute 10 · stage 19 · board 13 · map 13 · app 17 · portal 16 · theme 11 · layout 35+35 · phone 14 · doc 10	**Client frozen** Gemini path kept for Kannada only; the English process starts without the key	**Handoff** Notes, the model notice with CC-BY credits, the switch procedure
Factory & data 1 person · corpora, scripts, voices, alignment, filtering	**Fetch** Open Yap 1K requested 09:00; DailyTalk, AMI, MUSAN, RIRS fetched; consented Mitra voice clip recorded	**Pipeline** script → two voices → stereo WAV (Mitra left, user right) → aligner → moshi-finetune format; 10 dialogues	**500 scripts** Sakala conversations by gpt-oss-120b with interruptions, repairs, backchannels; voiced	**2,000 filtered** Canary + Whisper agree, WER < 10%; Open Yap subset converted on arrival	**Augment** MUSAN noise, room responses, phone band; round-1 manifest to CS	**Round 2 queue** 3,000 more hours; panel and suite failures turned into targeted scripts	**Hard cases** Far-field, crosstalk, interruptions at every point, numbers under noise	**Dataset frozen** Round-2 manifest; factory documented for the Kannada programme	**Archive** Datasets and manifests archived; licence credits compiled	**Carry on** Nightly factory cron left running for Kannada
Harness & evaluation 1 person · virtual mic, atlas, suites, FDB, panels, scorecard	**Atlas probes** Pause, interruption at 20 points, id reading, tool wait and lead phrase, noise; 10 repeats each	**Virtual mic** At the WebSocket with ms traces; atlas measured on Gemini and signed as the reference	**FDB baseline** Full-Duplex-Bench v1 on stock PersonaPlex; suites' fake Live pointed at the core	**First suites** relay 15 · live-UI 10 · turns 24 · nav 22 on the core; failures bucketed core / relay / harness	**Banks** bank 113 · guardrails 59 on the core; deviations file reviewed line by line	**Atlas + panel** Atlas on the core; 5-listener panel, 6 conversations; FDB v1 and v1.5 on round-1 weights	**Stress families** Noise, far-field, crosstalk, interruption-at-every-point added; load traces plotted	**Tune + recruit** Barge-in budget and pause tolerance from the atlas; 20 listeners recruited; 12 conversations recorded on both	**Scorecard** 20-listener blind A/B scored; every definition-of-done row with its measured value	**Sign-off sheet** Filled from the evidence folder; demo script walked twice
GPU 2 at night 20:00 → 07:00	downloads, hashes	factory: 10 dialogues	factory: 500 scripts voiced	factory: 2,000 filtered	**LoRA round 1 trains**	factory: round-2 synthesis	factory: hard cases	**LoRA round 2 trains**	idle, weights frozen	backups; cron for Kannada
Done when	Every model loads on the right card; atlas list signed; licence read answered	A person talks to the core on the web desk and hears it answer; Gemini atlas file filled	"Open cases in Hunsur" → function call → number spoken by the core; FDB baseline in evidence	**Gate A signed**: sovereign turn end to end, first word < 0.6 s, id equals the result	Trainer running with a manifest; bank and guardrails results owned	**Gate B signed**: six suites green, FDB ≥ PersonaPlex, panel within 0.5	Sessions-per-card number and the soak result in evidence	Regression results in evidence; trainer running; panel material ready	**Gate C signed**: atlas in tolerance, panel within 0.33; scorecard with no empty cell	**Sprint owner signs** the sheet; anything unsigned listed with a date

WHAT DONE MEANS · THE THREE GATES · THE TWO CARDS · THE RISKS THAT MATTER

every number on the sign-off sheet points to a file in mitra/evidence/sprint27/

Definition of done, day 10

BEHAVIOUR	TARGET
First word	< 0.5 s median · < 0.8 s p95
Barge-in cut	< 0.25 s
Ids and numbers spoken	0 deviations
Existing English suites	all green
Full-Duplex-Bench v1 / v1.5	≥ PersonaPlex published
20-listener panel vs Gemini	within 0.33 on naturalness
Sessions per card, p99 tick held	≥ 50 on GPU 1
2-hour soak at 50	0 dropouts
Egress during the rehearsal	no Gemini host

Captions and charts as she speaks · document reply in her voice · read-back bar · notifications · mute and stop: as on Gemini today.

Out of scope, stated on day 0: Kannada on the core · Gemini's emotional range · Gemini's exact voice.

The three gates, each with a written fallback

Gate A · Thu 30 Oct 18:30 · a sovereign English turn end to end
Pass: a question on the web desk → brain → function → result spoken by the core in Mitra's voice while the mic stays open; first word < 0.6 s; the id equals the result. Signed on a recording.
If missed: same-voice fallback that evening: Kyutai TTS 1.6B on the same voice embedding streamed into the output while the core keeps listening. Full duplex kept; a small voice mismatch on facts.

Gate B · Sat 1 Nov 18:30 · suites green, FDB at PersonaPlex's level
Pass: relay 15, live-UI 10, turns 24, nav 22, bank 113, guardrails 59 green on round-1 weights; FDB v1 ≥ published PersonaPlex; 5-listener panel within 0.5 of Gemini.
If missed: the load test moves to day 7; round-2 data is cut to what the failures need.

Gate C · Tue 4 Nov 18:30 · atlas in tolerance, panel within a third of a point
Pass: every atlas behaviour inside its tolerance on round-2 weights; 20-listener blind A/B within 0.33 on naturalness, not below on correctness; FDB held.
If missed: the best weights are frozen anyway, the gap is written on the scorecard, the rehearsal goes ahead with the gap stated. A third round is a follow-on, not a slip.

The plan has no slack after day 8: day 9 is the only buffer. A missed gate is written the same evening with its fallback, never deferred.

The two H200s, by day and by night

GPU 1 · 141 GB · the duplex worker, day and night
core 16 | sessions 60 | headroom 61
PersonaPlex 16 GB · STT 2 GB + Mimi + VAD · 0.6 GB per session: 50 promised, 100–150 fit · never used for training · admission control refuses the session it cannot serve well.

GPU 2 · 141 GB · by day
gpt-oss-120b 65 | cache 20 | w2 16 | free 40
The brain answers guided JSON for the existing router; the second worker carries overflow sessions.

GPU 2 · 141 GB · by night (days 4 and 7)
gpt-oss-120b 65 | finetune 40 | tools 20 | paused
The brain stays up for script writing; moshi-finetune 40 GB peak; TTS, ASR and the aligner 20 GB; the second worker is paused at 20:00 and back at 07:00.

Ports and evidence
Core 8040 / 8041 · vLLM 8050 · Kyutai STT 8042 · Mitra 8030 unchanged · test instances 8031 / 8032 via `MITRA_LIVE_URL` · egress logged at the firewall · `mitra/evidence/sprint27/` holds every measurement · backups in `tools/backups/sprint27/` · 8000 and 8001 untouched.

Risks, in order, with the fallback

1. **Text forcing not clean.** Gate A decides; the same-voice TTS fallback is ready on day 2.
2. **Open Yap arrives late.** DailyTalk, AMI and synthetic data carry round 1.
3. **VRAM contention on GPU 2.** Train only at night with the worker paused; Qwen3-30B-A3B as the lighter brain on those nights.
4. **NVIDIA licence read.** Answered on day 0; Moshiko (CC-BY-4.0) is the base either way, one extra night if needed.
5. **"Perfect" drifts to Gemini's voice or emotion.** Out of scope on day 0; a dated follow-on, not a slipped sprint.
6. **Latency under load.** Promise the count at which the p99 tick holds; the 3B distillation is the post-sprint lever.
7. **Synthetic accent learnt.** Two voice families; real speech weighted up in round 2.

Rhythm
09:30 stand-up (15 min) · 18:30 evening check, gates judged here · 20:00–07:00 GPU 2 belongs to the trainer and the factory · a daily note: done, not done and why, decisions.

Nothing from Gemini is stored except the atlas numbers. No Gemini output enters a dataset. KPCL and RDPR stay read-only.

Day 3 a sovereign turn · Day 5 suites green · Day 8 atlas and panel · Day 9 rehearsal with the Gemini key removed · 100% sovereign on the English path